

ZAHRA BENSLIMANE

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Education

- 2023–present **PhD, Computer Science**, *Commissariat Energie Atomique (CEA), IAEM Doctoral School (University of Lorraine)*
Laboratory : Embedded AI
Dissertation: Neural networks for improving speech understanding in real-time embedded systems
- 2021–2023 **Master of Engineering, Smart Systems Engineering**, *Sorbonne University*
Machine learning, deep learning, audio and signal processing, computer vision, big data, optimisation.
- 2019–2021 **Bachelor of Engineering, Electronics, Electrical Engineering & Automation**, *Sorbonne University*
Analog and digital electronics, embedded systems and microprocessors, control theory, electromagnetism, microprocessors and FPGAs.

PhD Research

My PhD research focuses on embedded AI for speech processing, with a particular emphasis on low-latency and low-computational-cost multi-microphone speech enhancement for resource-constrained systems. This project is conducted in collaboration with LORIA and Institut Pasteur research laboratories, with the objective of developing efficient AI-based embedded hearing-assistance devices.

Work Experience

- 2024–2026 **Part-Time Teaching Assistant**
ENSTA / IUT of Cachan
 - Supervised tutorials and practical sessions in Python and C programming.
- 2021–2023 **Machine Learning Engineer**
Renault Group – Part-time during academic periods, full-time during university breaks
 - Framed complex manufacturing problems by defining project scope and factory requirements.
 - Developed and evaluated deep learning models for industrial visual inspection systems.
 - Collaborated with startups and external partners to integrate AI solutions into industrial manufacturing workflows.
 - Contributed to the development of an MLOps platform for large-scale deployment of AI solutions and model monitoring.

Relevant Skills

Programming: Python, C/C++, MATLAB, SQL

Deep Learning: TensorFlow, PyTorch, Keras, pandas, scikit-learn

Experience with MLPs, CNNs, RNNs, Autoencoders, Transformers and neural network compression methods including folding, pruning, quantization, knowledge distillation and code optimization for low latency and low computational cost.

Audio Processing: Time-frequency analysis, audio compression, beamforming, Time Difference of Arrival (TDOA) estimation for sound source localization; audio source separation, speech enhancement.

Embedded Systems: NVIDIA Jetson Nano, Microcontroller programming in C/C++

Electrical: Oscilloscopes and multimeters, AC/DC power supplies, soldering and wiring.

Soft Skills: Problem framing, technical communication, autonomy, teamwork, teaching, cross-functional collaboration.

Languages French and English: Fluent. Arabic and Amazigh: Native;

Academic projects

- 2023 Underwater Image Analysis for deep sea species detection, segmentation and tracking.
- 2022 Neural Machine Translation for English to French translation.
- 2022 Augmented reality for inserting 3D objects within videos.
- 2021 Wire-guided delivery robot: Designed different sensor circuits from scratch and controlled them using microcontrollers.